

Skewness

Introduction

Skewness describes the direction in which a distribution is stretched or has a longer tail.

There are three common types of distributions:

1. **Symmetrical Distribution** – Both sides are approximately balanced.
2. **Right-Skewed Distribution (Positive Skewness)** – The distribution has a longer tail toward the right because of unusually high values.
3. **Left-Skewed Distribution (Negative Skewness)** – The distribution has a longer tail toward the left because of unusually low values.

A useful rule for identifying skewness is:

- **Mean > Median** → **Generally right-skewed**
 - **Mean < Median** → **Generally left-skewed**
 - **Mean $\approx$ Median** → **Approximately symmetrical**
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Q1. Distribution of Daily Income of Food Delivery Agents

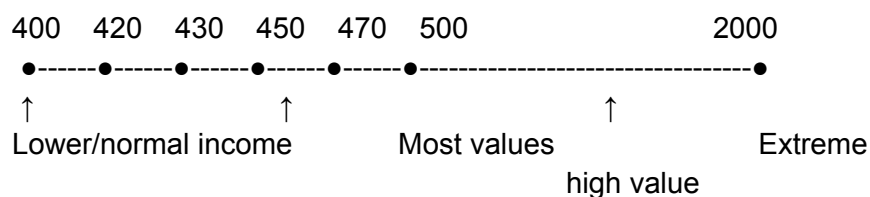
Given Data

Daily income of 7 food delivery agents:

400, 420, 430, 450, 470, 500, 2000

Number Line

Income (₹)



There is a large gap between ₹500 and ₹2,000.

Most of the observations are concentrated between ₹400 and ₹500, while one observation is much higher at ₹2,000.

Visual Interpretation

The distribution has a **longer tail toward the right** because of the unusually high income of ₹2,000.

Therefore:

The distribution is Right-Skewed\boxed{\text{The distribution is Right-Skewed}}

Reason

The value ₹2,000 is much larger than the other incomes and pulls the distribution toward the right. This creates a **positive/right skew**.

Q2. Mean, Median and Direction of Skewness

Given Data

350, 370, 400, 410, 415, 420, 2000350,\ 370,\ 400,\ 410,\ 415,\ 420,\ 2000

The data is already arranged in ascending order.

Step 1: Calculate the Mean

Formula:

Mean=Sum of all observationsNumber of observations\text{Mean}=\frac{\text{Sum of all observations}}{\text{Number of observations}}

Sum:

350+370+400+410+415+420+2000350+370+400+410+415+420+2000 =4365=4365

Number of observations:

n=7n=7

Therefore:

Mean=43657\text{Mean}=\frac{4365}{7} Mean=623.57\boxed{\text{Mean}=623.57}

So, the mean income is approximately ₹623.57.

Step 2: Calculate the Median

There are 7 observations.

For an odd number of observations, the median is the middle value.

The data is:

350, 370, 400, 410, 415, 420, 2000

The 4th value is the middle value.

Therefore:

Median = 410

Step 3: Compare Mean and Median

Measure	Value
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Mean	₹623.57
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Median	₹410
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We can see that:

$623.57 > 410$

Therefore:

Mean > Median

Conclusion

The distribution is **right-skewed (positively skewed)** because the mean (₹623.57) is considerably greater than the median (₹410). The unusually high income of ₹2,000 pulls the mean toward the right, while the median remains closer to the typical income of the group.

Q3. Monthly Pocket Money Dataset

Own Dataset

Suppose the monthly pocket money received by 8 friends is:

2000, 2200, 2500, 2500, 2600, 2700, 2800, 5000
2000, \ 2200, \ 2500, \ 2500, \ 2600, \ 2700, \ 2800, \ 5000

The data is already arranged in ascending order.

Step 1: Calculate the Mean

Formula:

Mean = $\frac{\text{Sum of all observations}}{\text{Number of observations}}$

Calculate the total:

2000+2200+2500+2500+2600+2700+2800+5000
2000+2200+2500+2500+2600+2700+2800+5000 = 22300

Number of observations:

$n=8$

Therefore:

Mean = $\frac{22300}{8}$ Mean = 2787.50
 $\boxed{\text{Mean} = 2787.50}$

So, the average monthly pocket money is **₹2,787.50**.

Step 2: Calculate the Median

There are 8 observations, so we have an **even number of observations**.

Therefore, we take the average of the two middle values.

The data is:

2000, 2200, 2500, 2500, 2600, 2700, 2800, 5000
2000, \ 2200, \ 2500, \ **2500**, \ **2600**, \ 2700, \ 2800, \ 5000

The two middle values are:

2500 and 2600
2500 \text{ and } 2600

Therefore:

$$\text{Median} = \frac{2500 + 2600}{2}$$

$$\text{Median} = 2550$$

Step 3: Determine the Skewness

Measure	Value
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Mean	₹2,787.50
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Median	₹2,550
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Since:

$$2787.50 > 2550$$

the mean is greater than the median.

Therefore:

The data is Right-Skewed

The **₹5,000** pocket money is considerably higher than the other values and pulls the mean upward.

What does this suggest?

The pocket money received by most friends is concentrated around **₹2,000–₹2,800**, while one friend receives substantially more. Therefore, the median of **₹2,550** may give a better idea of the typical pocket money than the mean.

Q4. Real-Life Example of a Skewed Dataset

Example: Instagram Followers

A real-life example of a skewed dataset is the **number of Instagram followers** of people in a group.

Suppose the follower counts are:

150, 220, 300, 450, 600, 900, 1500, 5000, 20000
150, \ 220, \ 300, \ 450, \ 600, \ 900, \ 1500, \ 5000, \ 20000

Most people have relatively small numbers of followers, while one or two accounts may have thousands or tens of thousands of followers.

Expected Distribution

This dataset would most likely be:

Right-Skewed \boxed{\text{Right-Skewed}}

Reason

The majority of users have a relatively small number of followers, while a small number of users may have **very large follower counts**. These unusually large values create a long tail on the right side of the distribution.

Real-Life Interpretation

This is why the **median number of followers** can sometimes give a better idea of the typical user's follower count than the mean. A few highly popular accounts can make the mean much larger than what an ordinary user typically has.